Capstone project – WIPRO DC Batch 5874

Jitendra Singh Tomar (Jeetu)

Note

1. Rename all the files with the naming convention as “Capstone-Project-1-<5-digit-ID.last_name>” as PDF document and upload it on the URL provided by RPS.
2. Upload only PDF documents with the specific name as mentioned above.
3. This document contains 6 projects in it, each in a different page.
4. **Its compulsory to attempt 3 projects.**

<u>Project Task 1: - "Create domain controller with NFS share"</u>

Objective: Create a domain controller using domain name as "wiprotraining.in" and create a shared NFS folder (C:\WiproB5874) on domain controller machine with read and write permissions and access this folder on the member machine (Node01 & Node02).

Setup:

- Use VMWare workstation 17 or above to create a windows server 2016 as domain controller with a custom LAN Segment names as "Wipro-network".
- Provide static IP as 192.168.10.10/24 with loopback DNS address & perform post-installation configuration.
- Install the required role and promote the server to domain controller.
- On DC, create a group using name "dcadmins" and add 2 users in it with the 'domain admins' permissions.
- Install another windows server 2019 (as GUI) with the name "Node01" and domain join these member/nodes to domain controller & verify.
- Create a new directory as "c:\WiproB5874" on domain controller with read and write permissions & list permission to a group called "dcadmins" and verify the access from the "Node01" machine.
- Create few files under "c:\ WiproB5874" and try accessing and modifying them on the member/node machine using the users created under "dcadmin" group.

Project deliverables:

- Document (take screenshot (image)) installation of both domain controller and Node01 machine.
- Document (take screenshot (image)) the post-installation setup & domain controller promotion.
- Document (take screenshot (image)) NFS sharing and accessing the data.

Project Task 2: - “Create LVM and create files within the same LVM.”

Objective: Install GUI CentOS 9 server. Attach 3 disks of 5GB each & create 2 LVMs from these disks of 15GB and mount at /lvm1 with EXT4 and /lvm2 with XFS file systems and verify by creating few files and directories in it.

Setup:

- Use VMWare workstation 17 or above to create a new CentOS 9**.**
- Change your linux server name to "LINUX-SVR-01".
- Change the IP address on linux to 192.168.10.20/24.
- Install the packages using YUM package manager.

Project deliverables:

- Create document (take screenshot (image)) installation of linux server.
- Create document (take screenshot (image)) of LVM creation.
- Create document (take screenshot (image)) of installation of packages using YUM command.

Project Task 3: - "Create storage space using 3 disks"

Objective:

Create a storage space using 3 disk each of 10GB (total 30GB) and access it on domain controller machine and verify.

Setup:

- Create storage space using 3 disks each of 10GB.
- Add appropriate virtual HDD and volume using "Simple" disk and thin provisioning.
- Collect appropriate screenshot (image) for the same.

Project deliverables:

- Create document (take screenshot (image)) for adding storage space.

<u>Project Task 4: - "Configure Windows Server Backup and store it in RAID 0 disk"</u>

Objective:

On Windows Server 2019, configure Windows Server Backup by installing appropriate feature and take backup of "**system state"** & "**C:\Users"** directory and store this backup in a newly created "RAID 0" volume with 2 disks of 10GB each (*create if RAID 0 isn't added*).

Setup:

- Install a Domain controller on Windows Server.
- Add 2 disks of 10GB each and configure RAID 0 after post installation configuration.
- Install Windows Server Backup (WSB) feature and configure it to take the backup of system's current state and users' directory i.e, C:\Users.

Project deliverables:

- Create document (take screenshot (image)) of configuring RAID 0.
- Create document (take screenshot (image)) of installing and configuring WSB and taking backup.

Project Task 5: - "Create a database and add the following entries using MySQL workbench."

Objective:

Perform the following operation. You are building a database for a city library to manage books and their availability.

Table name: books

Column Name: book_id, title, author, Genre, published_year, available_copies

1. CREATE

Write an SQL query to create the books table with appropriate data types and a primary key.

2. INSERT

Insert the following 4 books into the table:

```
book_id        title,author,genre,published_year, available_copies
101,The Alchemist,Paulo Coelho,Fiction,1993,5
102,Clean Code,Robert C. Martin,Tech,2008,2
103,Wings of Fire, , A. P. J. Abdul Kalam Biography,1999,3
104,Introduction to Algorithms, Thomas H. Cormen, Tech,2009,4
```

3. READ

Write a query to retrieve the titles and authors of all Tech books published after the year 2000.

4. UPDATE

One copy of "Clean Code" was returned. Increase its available_copies by 1.

5. DELETE

"The Alchemist" is no longer available. Remove it from the database.

Setup:

- Install MySQL server and access it using MySQL Workbench
- Create a database, within this database create a table with the data provided.
- Perform CRUD operation as instructed.
    - Adding bulk entries to the table
    - list the duplicates rows if any.
    - create index of the existing table.
    - perform:
        - truncate
        - drop
        - alter commands

Project deliverables:

- Create document (take screenshot (image)) of installing MySQL and MySQL Workbench.
- Create document (take screenshot (image)) of CRUD operation.

Project Task 6: - "Install NLB with two Nodes (Node01 & Node02)"

Objective:

Install and configure NLB using IIS web server sites on Node01 & Node02.

Setup:

- Install IIS webserver on Node01 & Node02.
- Install NLB on domain controller and configure High Availability.

Project deliverables:

- Create document (take screenshot (image)) of installing IIS Web server on Node01.
- Create document (take screenshot (image)) of installing IIS Web server on Node02.
- Create document (take screenshot (image)) of Install & configure NLB with proper screenshot (image).